

python equivalent of qnorm, qf and qch2 of R

Asked 10 years, 2 months ago Modified 2 years, 10 months ago Viewed 44k times 1 Part of R Language Collective

I need the quantile of some distributions in python. In R it is possible to compute these values using the qf, qnorm and qch2 functions. Is there any python equivalent of these R functions? I have been looking on scipy but I did not find anything.

python r scipy Edit tags

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asked Jul 11, 2014 at 10:03
answered Jul 11, 2014 at 10:03
17.3k 38 118 193

2 Answers

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You may find probability distributions in [scipy.stats](#). Every distribution defines a set of functions, for example if you go to [stats](#) distribution and scroll down the page you may find the methods which include

`ppf(q, loc=0, scale=1)` # Percent point function (inverse of cdf ~ percentiles)

that is what `scipy` calls Percent point function and does what `R` `q` functions does.
For example:

```
>>> from scipy.stats import norm
>>> norm.ppf(.5) # half of the mass is before zero
0.0
```

Same interface for other distributions, say `chi2`:

```
>>> from scipy.stats import chi2
>>> chi2.ppf(.5, df=2) # two degrees of freedom
3.218759248357215
```

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edited Nov 16, 2021 at 8:00 answered Jul 11, 2014 at 10:02
10.9k 209 4,563 8,253 77.8k 18 128 126

so percent point function compute the quantile of a distribution? - [Doroboi](#) Jul 11, 2014 at 10:38

Thanks for pointing out the oddly named Percent Point Function in scipy. Being unfamiliar with scipy, I've spent WAY too much time trying to google around was just about to ask how to do the equivalent of R's `qnorm` in Python when I landed on this page. - [D. Woods](#) Mar 30, 2017 at 21:56

Also want to point out that this is equivalent to Excel's `NORMINV()` function to help future searchers (Google). - [D. Woods](#) Mar 30, 2017 at 22:06

The equivalent of the R `qnorm()` function is: `scipy.stats.norm.cdf()` with python The equivalent of the R `qnorm()` function is: `scipy.stats.norm.ppf()` with python

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edited Mar 3, 2018 at 9:01 answered Feb 6, 2017 at 10:06
14.7k 25 74 103 10k 3 7

I think that answer may have aged. I can't seem to find `norm.ppf` in the docs, but I did find this [scipy.stats.norm.ppf](#) - [EngStudent](#) Aug 23, 2023 at 13:13

@EngStudent Here you can find `norm.ppf` [docs.scipy.org/doc/scipy/reference/generated/scipy.stats.norm.ppf.html](#) - [Hank Sarkar](#) Aug 29, 2023 at 20:40